

L2.3 How Derivatives Affect the Shape of a Graph

§3.3 — what you need to know

The derivatives table

① Find the **CNs** and the **PIPs**. ② Subdivide by **all of them together** — one line, not two. ③ Write the **evaluated** sign of f' and f'' on each piece. ④ **Draw** it: $\nearrow$ above, $\searrow$ below. ⑤ Sketch f underneath, at the same horizontal positions.

signs of f' , arrows, min/Max

$f'(x)$



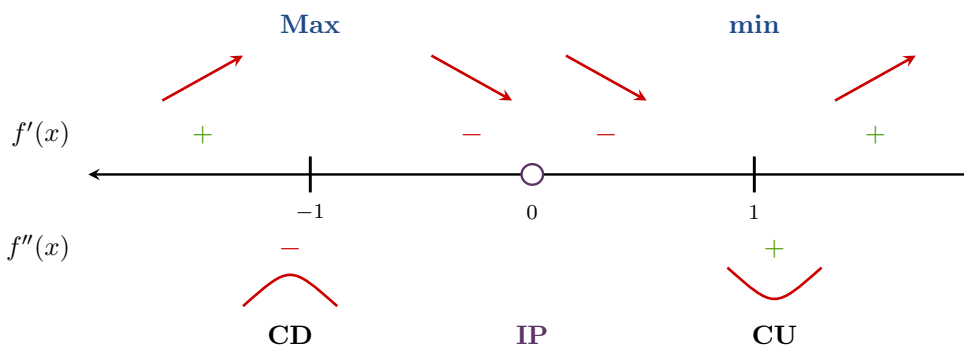
$f''(x)$

signs of f'' , cups, CD / CU, IP

sketch of f

CN critical number · **CP** critical point · **PIP** possible inflection point · **IP** inflection point · **CU** concave up · **CD** concave down · **HT** horizontal tangent · **@** at

THE TABLE FOR $f(x) = x^3 - 3x$



$f' = 3x^2 - 3$, so CP @ $-1, 1$. $f'' = 6x$, so $0 =$ possible IP — and the bend does change there, so it is an IP. Every sign was evaluated: $f'(-2) = 9$, $f'(0) = -3$, $f'(2) = 9$; $f''(-1) = -6$, $f''(1) = 6$.

THE THREE TESTS

I/D Test. If $f'(x) > 0$, $f \uparrow$. If $f'(x) < 0$, $f \downarrow$.

1st Derivative Test. At a critical number c — where $f'(c) = 0$ or $f'(c)$ DNE:

$f' : \oplus \rightarrow \ominus$ local Max $\ominus \rightarrow \oplus$ local min $\ominus \rightarrow \ominus$ or $\oplus \rightarrow \oplus$ no local extremum

“a critical number is a shortlist, not an answer”

2nd Derivative Test. CU if $f'' > 0$. CD if $f'' < 0$. IP if CU $\rightarrow$ CD or CD $\rightarrow$ CU.

$f'(c) = 0$ and CD $\Rightarrow$ local Max $f'(c) = 0$ and CU $\Rightarrow$ local min

“cup down”

It is **inconclusive** when $f''(c) = 0$, and unusable when $f''(c)$ DNE. Inconclusive means *no verdict*, not a verdict of *no extremum* — fall back to the sign switch. $f''(x) = 0$ or $f''(x)$ DNE makes x a **PIP** — a *possible* inflection point, which becomes an **IP** only once the sign check confirms the bend changed.

“possible is the whole word”

THE TWO CONDITIONS FOR A LOCAL EXTREMUM

For a local min or Max at a critical number c , **both** are required:

① the point must **exist** @ c ② f' must **switch sign** across c

$x^{1/3} + 3$ passes ① and fails ②. A hole or a vertical asymptote @ c fails ① however the signs behave.

WATCH OUT

~~$f'(c) = 0$ means an extremum @ $c \rightarrow$ it means candidate: the sign must switch, and the point must exist~~

~~x a critical number is where $f' = 0 \rightarrow$ or where f' DNE — as in $x^{1/3} + 3$ and $x^{2/3}$~~

~~$f'' = 0$ means inflection point $\rightarrow$ it means PIP. Confirm it with a sign change~~

~~no sign change, but I'll call it an IP $\rightarrow$ no change $\Rightarrow$ not an IP; none @ a CN $\Rightarrow$ not an extremum~~

~~the 2nd Derivative Test always works $\rightarrow$ inconclusive if $f''(c) = 0$, unusable if $f''(c)$ DNE~~

~~subdivide by the CNs, then by the PIPs $\rightarrow$ one line, subdivided by all of them together~~

~~concave down means decreasing $\rightarrow$ concavity is the bend, not the direction — f can rise while cupping down~~

~~I can see the sign $\rightarrow$ evaluate it, and write down the number you used~~